

Hyperspectral imaging compressed sensing based on spatial-spectral deep unfolding for agricultural products

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ABSTRACT

To address the challenge of balancing compressive-reconstruction performance, reconstruction speed, and interpretability in hyperspectral imaging (HSI) compressed sensing (CS) for agricultural products, this study proposes a spatial-spectral deep unfolding-based HSI CS model (SS-DUHCS). The model leverages the spatial-spectral characteristics of agricultural product HSI by unfolding the datacube row-wise into spatial-spectral slices for block-wise compressive measurement, enhancing compressive-reconstruction performance. A gradient-based optimization strategy embeds the physical characteristics of HSI from various agricultural products into the measurement matrix, improving adaptability to different datasets. During the reconstruction stage, a deep unfolding network transforms a traditional iterative optimization process into a multi-stage, physically interpretable network. This model learns the data characteristics of agricultural product HSI, achieving fast reconstruction while maintaining interpretability. A spatial and channel feature refinement module integrates into each unfolding-stage denoiser to extract effective features and eliminate redundancies, improving reconstruction quality while keeping the model lightweight. Experiments on four public and self-constructed agricultural HSI datasets demonstrate that SS-DUHCS consistently achieves superior reconstruction quality at low sampling rates (1%, 4%, and 10%) compared to iterative and deep learning-based baselines, with minimal model complexity. For instance, on the persimmons dataset at a 1% sampling rate, SS-DUHCS achieves 39.411 dB peak signal-to-noise ratio and 0.953 structural similarity index measure with just 30.69K parameters. The proposed measurement-reconstruction design generalizes well across various agricultural HSI data and is compatible with push-broom line-scan systems, supporting efficient non-destructive quality inspection in practical agricultural scenarios.

1. Introduction

Hyperspectral imaging (HSI) enables the simultaneous acquisition of spatial and spectral information from agricultural products, demonstrating significant potential for non-destructive inspection of both internal and external quality attributes [1,2]. However, current applications of HSI in agriculture are largely limited to laboratory environments, hindering large-scale deployment in grading, processing, and in-field monitoring. Unlike conventional RGB imaging, which captures three broad spectral bands, HSI records hundreds of continuous spectral channels for each spatial pixel across the visible to near-infrared range. Conse-

quently, a typical push-broom system scanning agricultural products at moderate spatial resolution (e.g., 1600×1600 pixels with 400 spectral bands) can generate several gigabytes of raw data per hyperspectral datacube. In high-throughput postharvest operations, such as packing houses, sorting and grading lines for fruits and vegetables, or grain inspection facilities handling thousands of samples per hour data volumes can reach terabyte scales daily, imposing significant burdens on data storage and transmission. Many agricultural applications rely on resource-constrained platforms, including in-field mobile systems, handheld instruments, and low-cost edge devices, for tasks such as on-site quality assessment and safety monitoring. These platforms often have

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limited storage capacity and network bandwidth, complicating the deployment of traditional full-sampling hyperspectral acquisition and processing pipelines [3].

To address these data volume and deployment challenges, previous studies have explored band selection and dimensionality reduction techniques. While these can reduce data size, they often discard potentially informative spectral details and are optimized for specific tasks, limiting flexibility across different quality indicators [4–6]. In contrast, a compressive-reconstruction strategy that performs compression at the acquisition stage and reconstructs the full hyperspectral datacube better aligns with agricultural HSI needs. This approach reduces data storage and transmission requirements while preserving original physical characteristics, allowing the reconstructed data to support multiple downstream, spectrum-based quality analysis tasks and benefiting food safety inspection and quality traceability. Compressed sensing (CS) aligns with this strategy by exploiting priors in appropriate transform domains to acquire fewer linear measurements than required by the Nyquist theorem, employing optimization algorithms or learning-based models for high-quality reconstruction [7,8]. Thus, developing a compressive-reconstruction CS method tailored to agricultural HSI is crucial for alleviating transmission and storage pressures while preserving full spectral information for versatile quality analysis, promoting the practical adoption of HSI-based inspection technologies in modern agriculture.

In existing research, CS methods for HSI of agricultural products have predominantly been model-based, utilizing prior data properties such as sparsity, correlation, low-rankness, and total variation for reconstruction [9–11]. Spectral information approaches have exploited the sparsity of plant HSI spectral data through greedy algorithms, enabling effective inversion of physicochemical parameters [12]. Enhanced performance has been achieved by combining adaptive guided filters with spectral dimension decorrelation techniques [13]. For spatial information, block-based CS methods employ discrete cosine transform sparse bases and random Gaussian matrices for sampling, with reconstruction via Stagewise Orthogonal Matching Pursuit (StOMP) [14]. Recursive sub-tensor methods also eliminate spatial background information, facilitating HSICS [15]. However, these methods typically focus on either the spatial or spectral dimension, limiting reconstruction performance. The iterative nature of model-based reconstruction processes often results in slower speeds, constraining practical utility.

Recently, data-driven deep learning approaches to CS have leveraged the strong representation capability of neural networks for rapid, high-precision reconstruction, significantly accelerating inference compared to purely iterative methods. Benefiting from their hierarchical feature extraction, convolutional neural networks (CNNs) jointly learn spatial structures and spectral correlations, overcoming the limitations of traditional methods that separate spatial and spectral dimensions. CNN-based models have performed end-to-end CS for visual images and remote-sensing HSI [16–18]. Transformer architectures, with global self-attention mechanisms, have enhanced hyperspectral image compression and representation learning by modeling long-range spatial-spectral dependencies [19]. Hybrid CNN-Transformer frameworks combine local convolutional modeling with global attention [20], surpassing traditional model-based CS in reconstruction quality at given sampling rates. However, they face limitations critical in agricultural HSI scenarios. Most networks are purely data-driven, lacking explicit physical model guidance and designed priors, making them less interpretable, highly dependent on training data distribution, and performance difficult to guarantee with limited training data or domain shifts [21]. Additionally, network architectures are often tightly coupled to the spatial and spectral dimensions of training data, limiting flexibility across different sensors and imaging configurations. The model sizes and computational complexity—especially for Transformer-based or CNN-Transformer hybrid models—are typically high, leading to substantial GPU memory and compute requirements. Some recent Transformer-based hyperspectral compression schemes utilize a heavy encoder and

a lightweight decoder [19], contrary to the requirements in agricultural HSI, where acquisition hardware should be lightweight. These characteristics hinder the deployment of purely deep-learning-based compression or CS methods on resource-constrained platforms commonly used in agricultural environments.

From a hardware perspective, existing compressive spectral imaging systems are primarily tailored to snapshot architectures, such as coded aperture snapshot spectral imaging (CASSI) and dual-camera compressive HSI, which utilize optical encoding to multiplex spatial and spectral information into a single measurement. These systems increase imaging speed by capturing multiple spectral channels within one exposure and have been extensively studied in laboratory settings. Recently, deep unfolding networks (DUNs) have emerged as a powerful reconstruction paradigm. A DUN reformulates traditional iterative optimization algorithms into a stacked neural network architecture, with each stage corresponding to iterations of the underlying solver, retaining clear physical meaning [22–28]. Compared to purely black-box deep models, DUNs embed the imaging forward model and handcrafted priors into the network structure, enhancing interpretability and stability while maintaining the fast inference of deep learning [29–33]. Several DUN-based frameworks have been developed for snapshot HSI reconstruction in CASSI and dual-camera systems, incorporating sophisticated spatial-spectral priors into the unrolled optimization process, achieving impressive reconstruction performance under realistic hardware degradations [34–36]. These methods are suitable for hardware designs that acquire hyperspectral information from a single exposure. However, snapshot systems often require specialized, precisely aligned optical components, making them less compatible with existing agricultural inspection lines, where retrofitting complex snapshot optics is costly and challenging. In contrast, agricultural hyperspectral inspection typically relies on push-broom line-scan imagers, which acquire data line by line as the sample or conveyor belt moves. Line-scan systems are widely deployed on industrial sorting lines and in-field inspection platforms, providing high spatial resolution and flexible field-of-view coverage with simple and robust hardware. Their main limitation is not the imaging speed, but rather the substantial data volume and bandwidth requirements associated with continuous scanning of numerous agricultural products. It is more practical to design a compressive sensing and reconstruction framework that seamlessly integrates with existing line-scan imagers, reducing data transmission and storage loads while preserving full spatial and spectral information for downstream analysis.

To this end, this study proposes a compressive measurement strategy integrating spatial-spectral dimensions and a DUN-based reconstruction approach for agricultural product HSI. In the compressive phase, the approach leverages the spatial texture characteristics and spectral correlations of agricultural product HSI data, unfolding the data row-wise into spatial-spectral slices for block-wise compressive measurement. Compared to traditional single-band spatial planes, these spatial-spectral slices maintain features across both dimensions, providing detailed and uniform information, thereby enhancing sparse representability and enabling effective compressive reconstruction. Additionally, a gradient-optimized measurement matrix is learned to capture features of diverse agricultural product HSI data, improving compressive performance. In the reconstruction phase, a DUN-based model maps the traditional iterative optimization process onto a deep neural network with multiple stages, where each stage corresponds to a single iterative update and carries explicit physical significance. This design combines the interpretability and prior modeling of optimization-based compressive sensing with the fast inference and strong representation capability of deep networks. Furthermore, a spatial and channel feature refinement module (SCFRM) is incorporated into the reconstruction framework to enhance reconstruction quality by extracting salient features and eliminating redundancy, while simultaneously reducing model parameters and computational load.

In summary, this study proposes a spatial-spectral deep unfolding HSI CS model (SS-DUHCS) that balances interpretability and efficiency

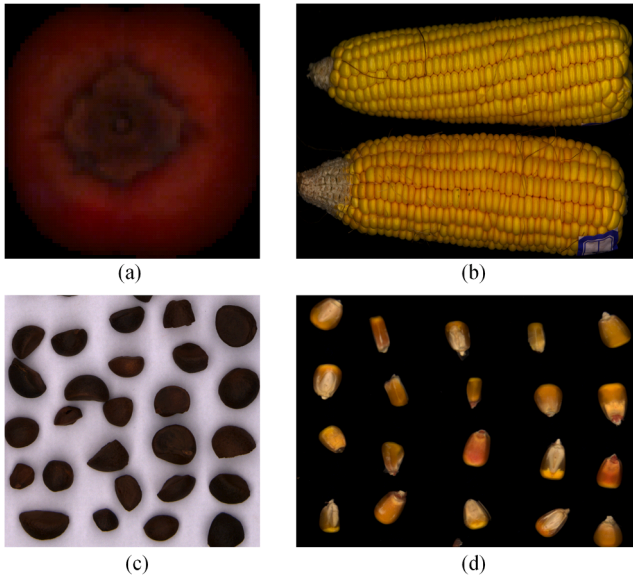


Fig. 1. RGB-visualized examples from the HSI datasets of four agricultural products. (a) Persimmons dataset, (b) Maize ears dataset, (c) Camellia oleifera seeds dataset, (d) Maize kernels dataset.

for rapid compressive measurement and reconstruction of agricultural product HSI data. The model has been validated using both open-source and self-constructed agricultural product datasets, demonstrating superior qualitative and quantitative performance, enhanced reconstruction quality, and reduced computational demands.

The paper is organized as follows: Section 2 describes the datasets and the SS-DUHCS model structure. Section 3 outlines the experimental setup and discusses the results. Section 4 concludes the article and highlights future directions.

2. Materials and methods

2.1. Data description

To evaluate the SS-DUHCS model's performance across diverse agricultural HSI characteristics, four representative agricultural product datasets with varying morphological and spectral properties were selected, as shown in Fig. 1 and detailed in Table 1. The selection criteria aimed to assess the model's robustness and generalization capability across different CS scenarios: (1) **Spatial scale diversity**: datasets included large fruits (persimmons, 50–80 mm diameter), medium-scale plant organs (maize ears, 200–300 mm length), and small kernels and seeds (maize kernels and camellia oleifera seeds, 8–15 mm diameter) to evaluate adaptability across spatial resolutions; (2) **Object density variation**: scenarios ranged from single-object imaging (persimmons) to multiple-object arrangements (maize ears with 2 specimens per datacube, scattered kernels/seeds with multiple specimens per datacube) to assess performance under varying spatial complexity; (3) **Background diversity**: varying background conditions included natural fruit surfaces, white reference panels, and black fabric backgrounds to test robustness against different imaging environments; and (4) **Data source heterogeneity**: integration of publicly available datasets (persimmons) and self-acquired datasets (maize ears, camellia oleifera seeds, maize kernels) using different HSI platforms to evaluate cross-platform generalization capability.

All datasets were acquired via push-broom hyperspectral imagers across the visible to near-infrared wavelength range (400–1000 nm). RGB images were generated by integrating reflectance values at 670 nm (red), 550 nm (green), and 450 nm (blue) from the full spectral range. The persimmons HSI data were sourced from a publicly available dataset

Table 1

Specifications of the collected agricultural products HSI data used in our experiments.

Hyperspectral imager	Wavelength range	Data set	Pixels per line
Specim FX10	400–1000nm (224 bands)	Persimmons	64
Resonon PIKA-XC2	400–1000nm (462 bands)	Maize ears Camellia oleifera seeds Maize kernels	1600

[37], while the HSI datasets for fresh edible maize ears, Camellia oleifera seeds, and maize kernels were acquired using a custom-built HSI platform based on the Resonon PIKA-XC2. Due to variations in data format and numerical range, normalization was applied to facilitate effective compressive reconstruction, scaling values to the range [0, 1].

$$I = \frac{I_o - I_{\min}}{I_{\max} - I_{\min}} \quad (1)$$

where I represents the normalized data, I_o represents the original data, $I_{\min}$ represents the minimum value among all pixels in the data, and $I_{\max}$ represents the maximum value among all pixels in the data.

2.2. Proposed method

The SS-DUHCS model consists of a compressive measurement phase and a reconstruction phase. The compressive measurement phase, implemented in the measurement module, compresses HSI data into an observation vector, facilitating efficient storage and transmission. The reconstruction phase, executed in the reconstruction module, employs a deep unfolding approach to recover HSI data from the observation vector while preserving information. The model's workflow is illustrated in Fig. 2. During the compressive measurement phase, original HSI data are unfolded row-wise into spatial-spectral slices and input into the measurement module. The data are segmented into blocks, vectorized, and subjected to compressive sampling using the measurement $\mathbf{A}$ to produce the observation vector $\mathbf{y}$. In the reconstruction phase, vector $\mathbf{y}$ is processed by an initialization module to restore its data structure. The DUN then utilizes K denoising modules to reconstruct the HSI datacube. To optimize HSI reconstruction quality, a loss function is introduced that simultaneously accounts for spatial and spectral information. The detailed principles and implementations of the measurement module, reconstruction module, and loss function are elaborated upon in the subsequent subsections.

2.2.1. Measurement module

The measurement module enables compressive measurement of HSI data for agricultural products. Unlike conventional end-to-end deep learning CS methods using convolutional encoder networks, this module employs a measurement matrix for HSI data compression and reconstruction, ensuring consistency between measurement and reconstruction. The structure of the measurement module is illustrated in Fig. 2. By utilizing the row-by-row scanning principle of push-broom HSI, each data row is unfolded into a spatial-spectral slice, encapsulating both spatial and spectral information [29]. This row-wise sampling facilitates real-time data transmission and processing. To manage hyperspectral data of varying dimensions while minimizing the measurement matrix size, each spatial-spectral slice X is partitioned into non-overlapping $n \times n$ blocks, a process referred to as Patching(X, n). This block decomposition enhances adaptability to different data sizes and reduces matrix dimensionality. The sampling procedure is formulated as follows:

$$\mathbf{y} = \mathbf{A} \times \text{vec}(\text{Patching}(X, n)) \quad (2)$$

where $\text{vec}(\cdot)$ denotes the vectorization operator, which reshapes an $n \times n$ data block into an $n^2 \times 1$ vector. $\mathbf{A}$ represents the measurement matrix with dimensions $m \times n^2$, and $\mathbf{y}$ is the $m \times 1$ observation vector. The compressive sampling rate is defined as the ratio of the data size of the observation vector $\mathbf{y}$ to that of the original vector during the compressive

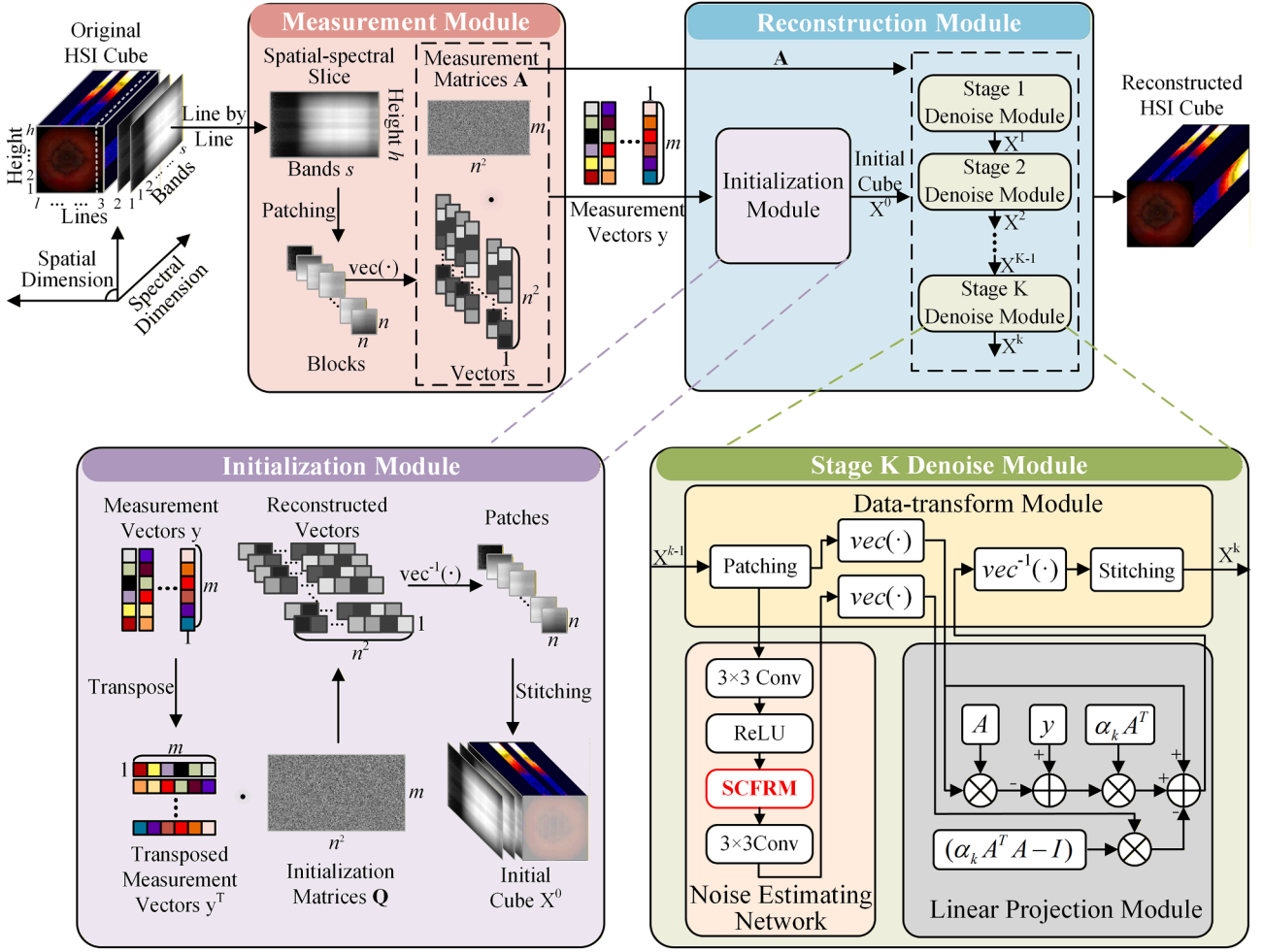


Fig. 2. The block diagram of the proposed SS-DUHCS model system architecture for HSI data CS.

sampling process, expressed as follows:

$$\text{CS ratio} = \frac{m \times 1}{n^2 \times 1} = \frac{m}{n^2} \quad (3)$$

Conventional HSI CS methods utilizing measurement matrices typically rely on fixed matrices, such as Gaussian random matrices. While these matrices offer broad applicability, they do not leverage the unique characteristics of imaged subjects. Agricultural products exhibit diverse morphologies, sizes, and surface materials, leading to variations in the spatial distribution and spectral properties of their hyperspectral data. To enhance the model's adaptability to diverse agricultural product data and improve information extraction efficiency, this study proposes a gradient optimization-based approach that incorporates the physical properties of HSI data from agricultural products into the measurement matrix. By optimizing the matrix, this method captures spatial distributions and spectral correlations more accurately. The initially random measurement matrix is designed as a trainable parameter matrix, adjustable via gradient optimization. A tailored loss function guides the iterative refinement of the matrix values during training to better align with the characteristics of agricultural product hyperspectral data. Training utilizes both open-source and self-constructed agricultural product datasets, optimizing the matrix to reflect the physical properties of various products. This enables effective compressive measurement and accurate reconstruction across multiple types of agricultural product data. The approach enhances information extraction precision and improves the model's generalization capability and applicability.

From a hardware perspective, the proposed measurement module is compatible with conventional push-broom line-scan hyperspectral imagers. Each scan line unfolds into a spatial-spectral slice, partitioned into non-overlapping blocks. Each vectorized block is multiplied by a learned measurement matrix $\mathbf{A}$, reducing to dense matrix-vector multiplications, efficiently implemented on low-power embedded platforms (e.g., FPGAs, embedded GPUs, or application-specific SoCs) integrated into the acquisition side of a line-scan system. In practice, block-wise sampling synchronizes with the line-by-line scanning process: once a line is acquired, it is segmented into blocks, compressed by $\mathbf{A}$, and the resulting low-dimensional measurements are transmitted and stored instead of the full-resolution hyperspectral data. This design preserves the optical path of the push-broom imager and adds only a lightweight electronic module for matrix-vector products, enabling seamless integration into existing hardware without the complex snapshot optics required in CASSI-like systems. Full hyperspectral datacubes can be selectively reconstructed on more powerful servers or cloud platforms when required for downstream analysis tasks.

2.2.2. Reconstruction module

The central principle of reconstruction in model-based CS algorithms is to recover the original signal $\mathbf{x}$ by optimizing the objective function. The reconstructed signal $\hat{\mathbf{x}}$ is formulated as follows:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} D(\mathbf{y}, \mathbf{A}\mathbf{x}) + \lambda \mathcal{R}(\mathbf{x}) \quad (4)$$

where $D(\mathbf{y}, \mathbf{A}\mathbf{x})$ represents the data fidelity term (e.g., l_1 or l_2 norms), ensuring the reconstructed signal aligns with the observed data, and

$\mathcal{R}$ denotes the regularization term (e.g., an l_1 norm-based sparsity constraint), incorporating prior knowledge. Building on this formulation, traditional approaches commonly employ an iterative optimization method to address this problem, notably the Approximate Message Passing algorithm, which converges to the optimal solution through the following steps [38]:

$$x_k = \eta(x_{k-1} + \mathbf{A}^T(y - \mathbf{A}x_{k-1})) \quad (5)$$

where x_k and x_{k-1} denote the reconstructed signals at the k -th and $k-1$ -th iterations, respectively, $\mathbf{A}^T$ represents the transpose of the measurement matrix $\mathbf{A}$, and $\eta(\cdot)$ is a predetermined denoising function (e.g., a soft thresholding function) based on prior knowledge of the data. From the measurement equation $y = \mathbf{A}x$, substituting into Eq. (5) yields:

$$x_k = \eta(x + (\mathbf{A}^T \mathbf{A} - \mathbf{I})(x - x_{k-1})) \quad (6)$$

where $\mathbf{I}$ denotes the identity matrix, and the error term is defined as $e_k = (\mathbf{A}^T \mathbf{A} - \mathbf{I})(x - x_{k-1})$. This simplifies to:

$$x_k = \eta(x + e_k) \quad (7)$$

The original signal x can be progressively approximated by iteratively minimizing the error term e_k , a process interpretable as signal denoising. However, practical computations often encounter significant computational burdens and slow convergence rates. Recent studies have demonstrated that the DUN method reformulates the traditional iterative optimization process into a stacked architecture of deep neural network modules, significantly enhancing reconstruction speed and providing clear physical interpretability [23]. Drawing inspiration from this approach, this study transforms the computationally intensive iterative process into a deep network comprising multiple stages, each corresponding to a single iterative update and retaining distinct physical significance. The proposed reconstruction module, illustrated in Fig. 2, comprises an initialization module and a denoising module.

The initialization module transforms the observation vector y into an HSI datacube with dimensions matching the original data, providing an effective initial estimate. To achieve this, a trainable initialization matrix $\mathbf{Q}$ is designed, with its initial values optionally aligned with the measurement matrix $\mathbf{A}$. The data restoration process employs vector reshaping and block concatenation operations, as depicted in Fig. 2. The initialization procedure is represented as follows:

$$x^0 = \text{Stitching}(\text{vec}^{-1}(y^T) \mathbf{Q}, n) \quad (8)$$

where $\text{Stitching}(\cdot)$ and $\text{vec}^{-1}(\cdot)$ denote block concatenation and inverse vectorization, respectively, restoring the data to its original dimensions.

From the definition of the error term e_k , it is evident that when $\mathbf{A}$ remains constant, e_k depends solely on x and x_{k-1} . Thus, a deep neural network $N_k(\cdot)$ can estimate $x - x_{k-1}$, enhancing resilience to the complex noise inherent in HSI data. To replace the iterative process, k stages of denoising modules are constructed, with each stage improving reconstruction. Simultaneously, a trainable step parameter α_k is introduced for greater flexibility. Based on Eq. (5) and Eq. (6), the k -th stage denoising module, derived via linear unfolding, is represented as follows:

$$x_k = \alpha_k \mathbf{A}^T(y - \mathbf{A}x_{k-1}) + x_{k-1} - (\alpha_k \mathbf{A}^T \mathbf{A} - \mathbf{I})\text{vec}(N_k(x_{k-1})) \quad (9)$$

where the first term $\alpha_k \mathbf{A}^T(y - \mathbf{A}x_{k-1})$ represents the gradient descent term, adjusting the signal estimate based on the observation vector y ; the second term x_{k-1} is the retained estimate from the prior iteration, preventing over-updating; and the third term $(\alpha_k \mathbf{A}^T \mathbf{A} - \mathbf{I})\text{vec}(N_k(x_{k-1}))$ provides adaptive denoising via the neural network.

Fig. 2 illustrates the k -th stage denoising module, which comprises three main components: a data conversion module, a linear mapping module, and a denoising network. The data conversion module performs block segmentation and vectorization without introducing additional floating-point computations. The linear mapping module implements the linear computations specified in Eq. (9). The denoising network, realizing $N_k(\cdot)$, is constructed from a combination of 3×3 convolutions with ReLU activation functions and the SCFRM proposed in this study.

The design of SCFRM addresses three challenges in agricultural HSI CS reconstruction: (1) **Spectral-spatial coupling**: HSI shows strong inter-band correlations due to spectral continuity, necessitating the refinement module to consider spatial structures and inter-channel dependencies jointly. (2) **Computational efficiency**: Practical agricultural deployment relies on resource-constrained devices, requiring low parameter and FLOP overhead for refinement. (3) **Stage-wise stability under severe undersampling**: In deep unfolding, the denoiser is repeatedly applied across stages; thus, feature refinement should suppress artifacts and stabilize intermediate estimates without over-aggressive enhancement.

The structure of SCFRM is shown in Fig. 3. SCFRM is designed as a lightweight spatial-channel feature refinement module for the denoising sub-network in each unfolding stage, comprising two tightly coupled components: an Adaptive Spatial Feature Refinement Module (ASFRM) and a Channel Feature Refinement Module (CFRM).

ASFRM

ASFRM employs a dual-branch architecture to separate information-rich features from information-poor ones through a lightweight element-wise gating mechanism. Let $F \in \mathbb{R}^{B \times C \times H \times W}$ denote the input feature map, where B is the batch size, C is the channel number, and H and W are the spatial height and width, respectively. ASFRM first applies group normalization to obtain normalized features $\tilde{F}$:

$$\tilde{F} = \text{GN}(F) \quad (10)$$

where $\text{GN}(\cdot)$ is group normalization with a learnable channel-wise scale vector $\gamma = [\gamma_1, \gamma_2, \dots, \gamma_C] \in \mathbb{R}^C$. Here, γ_c denotes the scale parameter of the c -th channel. To build a gating signal without introducing additional convolutional layers, γ is normalized and reshaped to $v \in \mathbb{R}^{1 \times C \times 1 \times 1}$ for broadcast multiplication.

$$v = \frac{\gamma}{\sum_{c=1}^C \gamma_c} \quad (11)$$

The element-wise gating map $R \in \mathbb{R}^{B \times C \times H \times W}$ is then computed by:

$$R = \sigma(\tilde{F} \odot v) \quad (12)$$

where $\sigma(\cdot)$ denotes the sigmoid function and $\odot$ denotes element-wise multiplication. The gating map R provides a spatially varying indicator of feature saliency, which is used to split F into two complementary branches, allowing CS artifacts to be suppressed and stabilizing stage-wise denoising. A fixed gate threshold $t \in (0, 1)$ (set to $t = 0.5$ in our implementation) is used to construct two masks:

$$M_1(u) = \begin{cases} 1, & R(u) > t, \\ R(u), & R(u) \leq t, \end{cases} \quad M_2(u) = \begin{cases} 0, & R(u) > t, \\ R(u), & R(u) \leq t, \end{cases} \quad (13)$$

where u indexes each element in the feature tensor. The two gated branches are:

$$F_1 = M_1 \odot F, \quad F_2 = M_2 \odot F \quad (14)$$

which can be interpreted as emphasizing information-rich responses (F_1) while retaining complementary responses (F_2). To promote information exchange between the two branches with negligible cost, ASFRM performs a cross-branch reconstruction by splitting each branch into two channel halves:

$$F_1 = [F_{1a}, F_{1b}], \quad F_2 = [F_{2a}, F_{2b}] \quad (15)$$

where $F_{1a}, F_{1b}, F_{2a}, F_{2b} \in \mathbb{R}^{B \times (C/2) \times H \times W}$. The ASFRM output is then formed by cross addition and concatenation:

$$F_{AS} = [F_{1a} + F_{2b}, F_{1b} + F_{2a}] \quad (16)$$

Compared with conventional attention blocks that mainly reweight a single feature stream, ASFRM explicitly constructs two complementary streams and reconstructs them via cross-branch aggregation, enabling adaptive refinement based on feature informativeness and improving the stability of stage-wise denoising.

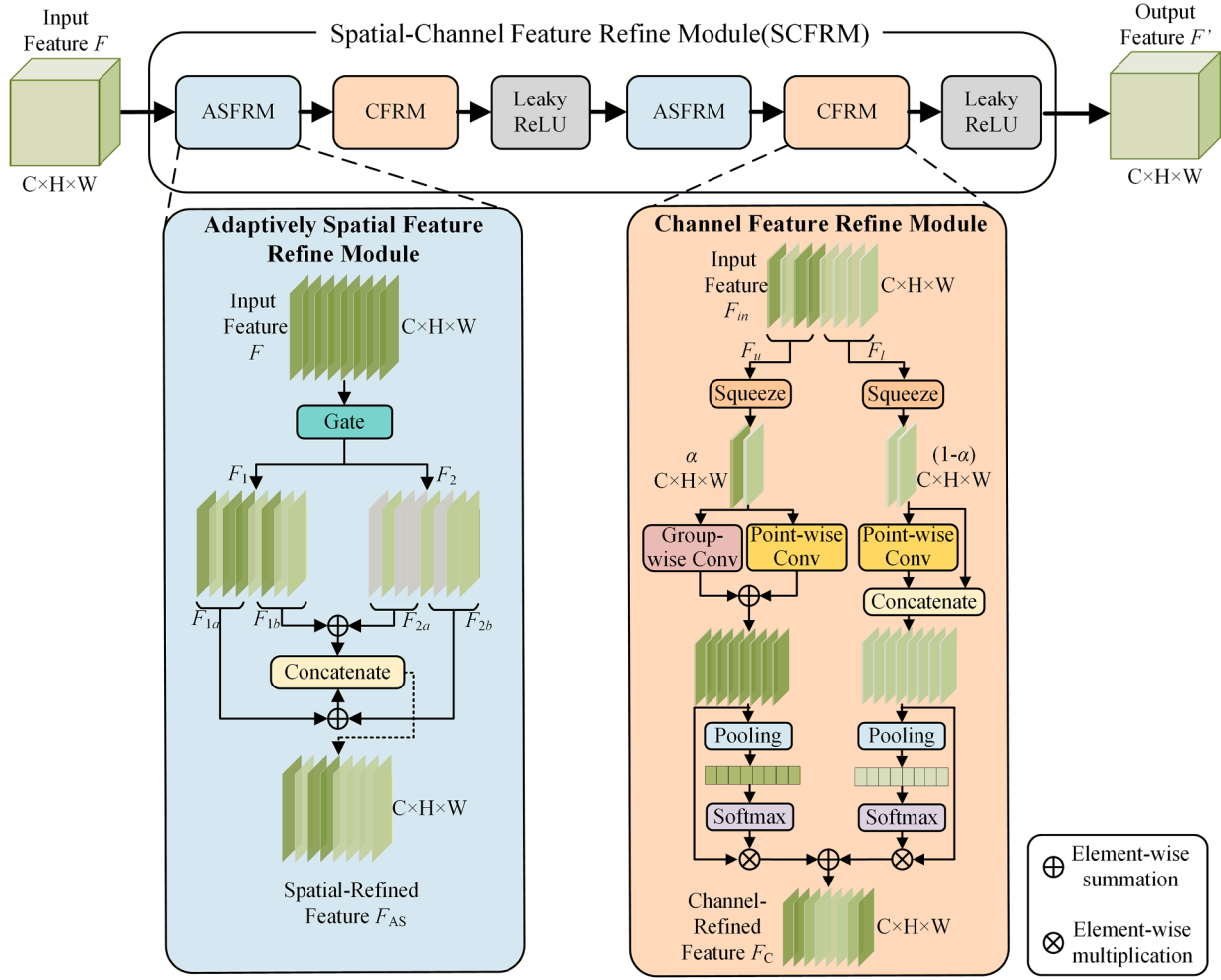


Fig. 3. Network architecture of the SCFRM.

CFRM

CFRM refines features along the channel dimension using a split-transform-merge pipeline. It (i) splits channels into upper and lower groups, (ii) applies distinct low-cost transforms to each group to capture complementary information, and (iii) computes a channel-wise attention weight via global pooling to reweight the merged features.

Let $F_{in} \in \mathbb{R}^{B \times C \times H \times W}$ denote the input to CFM. Let $\alpha \in (0, 1)$ be the channel split ratio, and define $C_u = \lfloor \alpha C \rfloor$ and $C_l = C - C_u$. CFM first splits F_{in} along the channel dimension:

$$F_{in} = [F_u, F_l] \quad (17)$$

where $F_u \in \mathbb{R}^{B \times C_u \times H \times W}$ and $F_l \in \mathbb{R}^{B \times C_l \times H \times W}$.

Next, both branches apply a 1×1 “squeeze” convolution to reduce channel dimensionality. After squeezing, the upper branch applies a group-wise convolution and a point-wise convolution in parallel *in parallel*, summing them to enhance local spatial structures with low overhead. The lower branch employs a lightweight point-wise transform, which is concatenated with a skip connection (i.e., the transformed feature is concatenated with the identity feature) to preserve complementary information. Following these transforms (including the necessary point-wise mappings), both branches are mapped to the same channel dimension and denoted as:

$$U \in \mathbb{R}^{B \times C \times H \times W}, \quad L \in \mathbb{R}^{B \times C \times H \times W} \quad (18)$$

To adaptively emphasize informative channels, CFM computes a channel-wise attention weight for each branch using global average

pooling and softmax:

$$w_u = \text{Softmax}(\text{GAP}(U)), \quad w_l = \text{Softmax}(\text{GAP}(L)) \quad (19)$$

where $\text{GAP}(\cdot)$ denotes global average pooling that maps a feature map to a tensor of size $B \times C \times 1 \times 1$, $\text{Softmax}(\cdot)$ is applied along the channel dimension, and $w_u, w_l \in \mathbb{R}^{B \times C \times 1 \times 1}$ are the resulting channel-wise weights.

Finally, each branch is reweighted and the refined output $F_C \in \mathbb{R}^{B \times C \times H \times W}$ is obtained through element-wise summation:

$$F_C = (w_u \odot U) + (w_l \odot L) \quad (20)$$

Overall, SCFRM refines spatial responses via gated cross-branch reconstruction (ASFRM) and then performs efficient channel refinement (CFRM). When inserted into each unfolding-stage denoiser, SCFRM enhances reconstruction quality while maintaining a lightweight model, which is crucial for patch-wise HSI CS reconstruction under strict computational and memory constraints.

2.2.3. Loss function

To evaluate the reconstruction quality of HSI and enable gradient optimization of network model parameters, an effective loss function must be designed. The L_2 and L_1 losses are the most widely used in contemporary research. The L_2 loss, which computes mean squared error (MSE), offers rapid convergence but is prone to gradient explosion with outliers. The L_1 loss, based on the mean absolute error, mitigates sensitivity to outliers; however, its non-differentiability at zero can induce oscillations near the optimal solution, hindering convergence. The smooth L_1 loss addresses these shortcomings, facilitating stable and rapid convergence

to the optimal solution [39]. For original hyperspectral data $f_{or}^{(b)}(s, h, l)$ and reconstructed data $f_{re}^{(b)}(s, h, l)$, with a batch size B , height H , number of spectral channels S , and number of scan lines L , the Smooth L_1 loss is expressed as follows:

$$\text{Loss}_1 = \frac{1}{BLHS} \sum_{b=1}^B \sum_{l=1}^L \sum_{h=1}^H \sum_{s=1}^S \text{smooth}_{L_1}(f_{re}^{(b)}(s, h, l) - f_{or}^{(b)}(s, h, l)) \quad (21)$$

where

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases} \quad (22)$$

The aforementioned loss functions are well-suited for conventional visual image tasks. However, HSI data, unlike standard visual images, encompass rich spectral information. To ensure both spatial and spectral fidelity in the reconstructed data, this study introduces a Spatial-Spectral Total Variation (SSTV) regularization term. This term leverages the Total Variation model to jointly account for spatial and spectral correlations [40]. SSTV is incorporated into the loss function to penalize gradients in each reconstructed spatial-spectral slice $f_{re}^{(b)}(:, :, l)$ across spatial and spectral dimensions, expressed as:

$$\text{Loss}_2 = \frac{1}{BL} \sum_{b=1}^B \sum_{l=1}^L R_{\text{SSTV}}(f_{re}^{(b)}(:, :, l)) \quad (23)$$

where

$$R_{\text{SSTV}}(X) = \|\nabla_h X\|_1 + \|\nabla_s X\|_1 \quad (24)$$

where ∇_h and ∇_s denote the finite-difference operators along the spatial and spectral directions of a spatial-spectral slice X , respectively, and $\|\cdot\|_1$ denotes the ℓ_1 norm (i.e., the sum of absolute values of all elements).

In summary, the loss function for this model $\text{Loss}_{\text{total}}$ is constructed by combining the SSTV regularization term with the smooth L_1 loss, formulated as:

$$\text{Loss}_{\text{total}} = \text{Loss}_1 + \lambda \text{Loss}_2 \quad (25)$$

where λ serves as a hyperparameter to balance the contribution of the SSTV regularization term.

3. Experiment and results analysis

3.1. Experimental settings

The data processing platform for this study includes an Intel Xeon Gold 6430 processor, an Nvidia RTX4090 24GB graphics card, and 120GB of memory. The operating system is Ubuntu 22.04, with CUDA version 11.8, Python version 3.10, and PyTorch version 1.13.

The experimental datasets were partitioned as follows: 10% of samples from the persimmons dataset served as the test set, while the remainder was used for training. For the maize kernels, camellia oleifera seeds, and maize ears datasets, 10% of consecutive rows were randomly selected as the test set, with the remaining data forming the training set.

During network training, the AdamW optimizer was utilized with an initial learning rate of 0.001. The training phase consisted of 50 epochs to ensure model convergence. Data blocks of size 33×33 were treated as individual batches, with a batch size of 128 to optimize memory usage and training efficiency. The random seed was set to 114,514 to ensure reproducibility of the results. Based on the model architecture and input data characteristics, the number of deep unfolding stages K was set to 7, and the loss function weight was tuned via validation to 0.01.

3.2. Evaluation metrics

The experimental results are evaluated using four evaluation metrics: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) [41], root mean square error (RMSE), and spectral angle mapper (SAM) [42]. Definitions of these metrics are as follows:

1) PSNR evaluates the quality of signal reconstruction, expressed in decibels (dB). A higher PSNR value indicates a better quality of the reconstructed HSI signal. PSNR is calculated using the formula:

$$\text{PSNR} = 10 \log_{10} \left(\frac{R^2}{\text{MSE}} \right) \quad (26)$$

where R denotes the maximum signal value, and MSE is the mean squared error between the original and reconstructed HSI data, defined as:

$$\text{MSE} = \frac{1}{LHS} \sum_{l=1}^L \sum_{h=1}^H \sum_{s=1}^S [f_{re}(s, h, l) - f_{or}(s, h, l)]^2 \quad (27)$$

where H denotes the height of the HSI, the number of spectral channels as S , and the number of scan lines as L . $f_{re}(s, h, l)$ and $f_{or}(s, h, l)$ represent the pixel intensities at position (s, h, l) in the original and reconstructed HSI, respectively.

2) SSIM assesses spatial reconstruction quality based on luminance, contrast, and structure. SSIM values range from 0 to 1, with higher values indicating greater similarity. The formula for SSIM is as follows:

$$\text{SSIM} = \frac{1}{S} \sum_{s=1}^S \frac{(2\mu_1\mu_2 + c_1)(2\sigma_{12} + c_2)}{(\mu_1^2 + \mu_2^2 + c_1)(\sigma_1^2 + \sigma_2^2 + c_2)} \quad (28)$$

where μ_1 and μ_2 are the pixel means, σ_1^2 and σ_2^2 are the variances of the reconstructed and original images for the s -th spectral channel, respectively, and σ_{12} is their covariance. The constants $c_1 = (K_1 D)^2$ and $c_2 = (K_2 D)^2$, with $K_1 = 0.01$, $K_2 = 0.03$, and $D = 1.0$ (the dynamic range of spatial pixel values), stabilize the division to prevent near-zero denominators.

3) RMSE measures reconstruction accuracy, calculated as:

$$\text{RMSE} = \sqrt{\text{MSE}} \quad (29)$$

4) SAM, expressed in degrees, evaluates spectral domain reconstruction performance by computing the angle between spectral vectors of the original and reconstructed HSI data at each spatial position. Smaller angles indicate greater spectral similarity. It is defined as:

$$\text{SAM} = \frac{1}{LH} \sum_{l=1}^L \sum_{h=1}^H \arccos \left(\frac{f_{or}(:, h, l) \cdot f_{re}(:, h, l)}{\|f_{or}(:, h, l)\|_2 \times \|f_{re}(:, h, l)\|_2} \right) \quad (30)$$

where $f_{re}(:, h, l)$ and $f_{or}(:, h, l)$ are the spectral vectors across all channels at position (h, l) for the reconstructed and original HSI, respectively, $\cdot$ denotes the dot product, and $\|\cdot\|_2$ represents the ℓ_2 -norm.

3.3. Results

3.3.1. Quantitative and qualitative results

The proposed SS-DUHCS model is compared with six hyperspectral compression methods known for strong reconstruction performance: two model-based iterative methods with three-dimensional total variation regularization (JTEnRe3DTV [43] and E3DTV [44]) and four deep learning-based approaches. Among the deep models, DCSN [17] and BTC-Net [45] are CNN-based CS networks, CTCN is a CNN-Transformer hybrid model [20], and HyCoT [19] is a Transformer-based autoencoder with a heavy encoder and a lightweight decoder. The denoising module of the DUN comprises seven stages, corresponding to seven iterations in traditional methods. The iterative methods JTEnRe3DTV and E3DTV are tested with 7 and 70 iterations, respectively. These methods reconstruct the persimmons dataset at compressive sampling rates of 1%, 4%, and 10%. Reconstruction performance is evaluated from both quantitative and qualitative perspectives.

Table 2 and Fig. 4 summarize the reconstruction performance of all methods on the persimmons dataset at sampling rates of 1%, 4%, and 10%. SS-DUHCS achieves the best reconstruction quality across these rates while maintaining a smaller model scale and favorable computational efficiency. Model-based iterative methods JTEnRe3DTV and

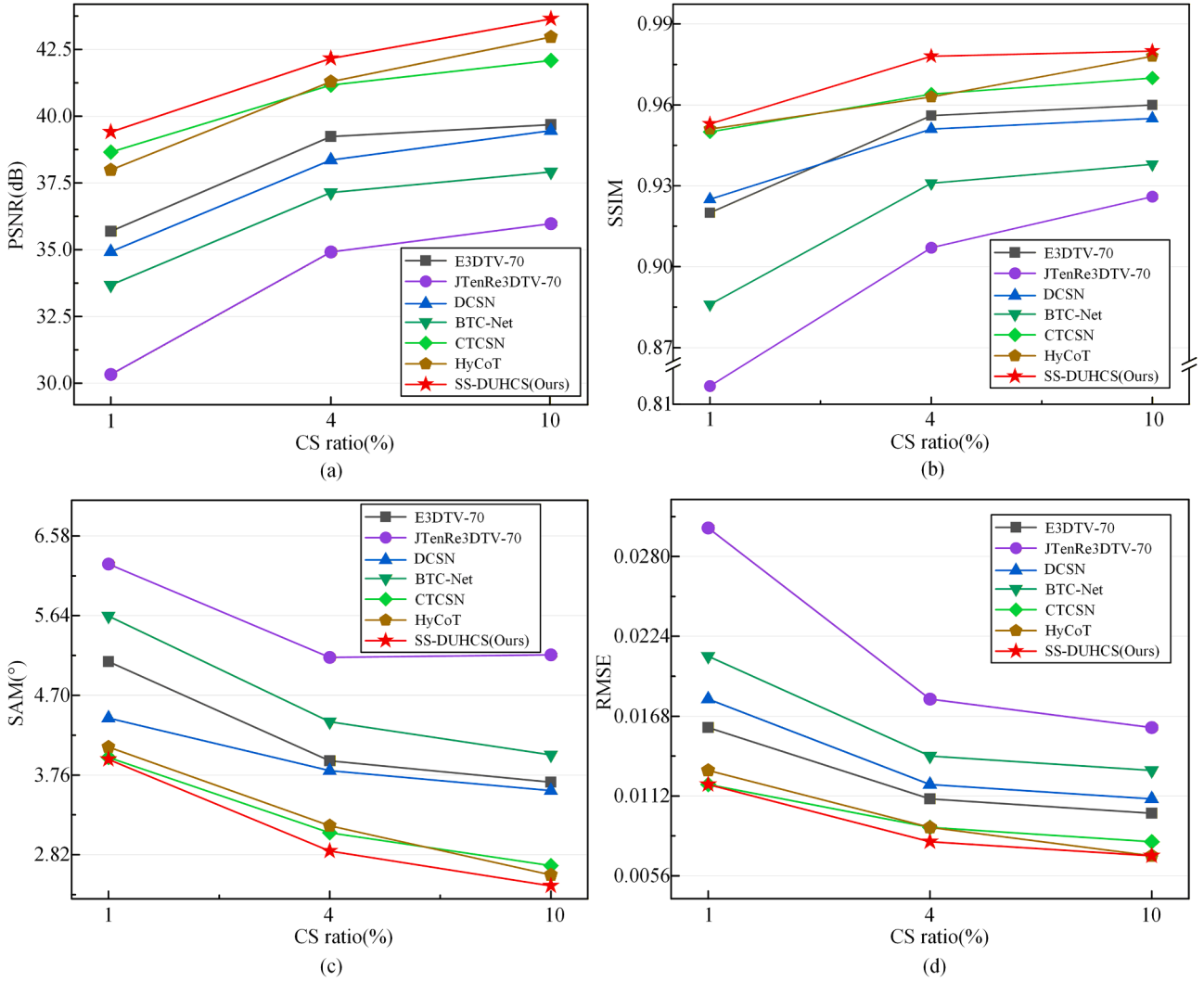


Fig. 4. Performance comparison of our model and competitive HSI CS methods at different sampling rates (1%, 4%, and 10%) on the metrics. (a) PSNR, (b) SSIM, (c) SAM, (d) RMSE.

E3DTV, with only 7 iterations, underperform at all sampling rates, reflecting the limitations of low-iteration optimization. Increasing the iteration count to 70 significantly improves PSNR and SSIM; however, the runtimes (hundreds of seconds per cube) are prohibitive for practical use. CNN-based CS models, DCSN and BTC-Net, outperform the iterative baselines in PSNR and SSIM at all sampling rates, but still lag behind SS-DUHCS, particularly in SAM and RMSE, which are critical for preserving spectral fidelity in agricultural HSI. Recent Transformer and CNN-Transformer hybrid models, HyCoT and CTCN, enhance reconstruction quality at 4% and 10% sampling rates and exhibit strong PSNR/SSIM performance. However, SS-DUHCS maintains the highest PSNR and SSIM and the lowest RMSE and SAM at all sampling rates. For instance, at a 1% sampling rate, SS-DUHCS achieves 39.411 dB PSNR and 0.953 SSIM, outperforming all other methods. At a 10% sampling rate, it achieves 43.649 dB PSNR and 0.980 SSIM, with the lowest SAM (2.454°), indicating superior spatial and spectral reconstruction.

Regarding model scale and computational efficiency in Table 2, we compare only deep learning-based approaches (DCSN, BTC-Net, HyCoT, CTCN, and SS-DUHCS), as iterative JTenRe3DTV and E3DTV lack fixed parameter counts or FLOPs. For a fair comparison, all FLOPs in Table 2 are defined as the *total* number of floating-point operations needed to process one hyperspectral cube of size $64 \times 64 \times 224$, summing the patch- or block-based forward passes (full-cube input for Hy-

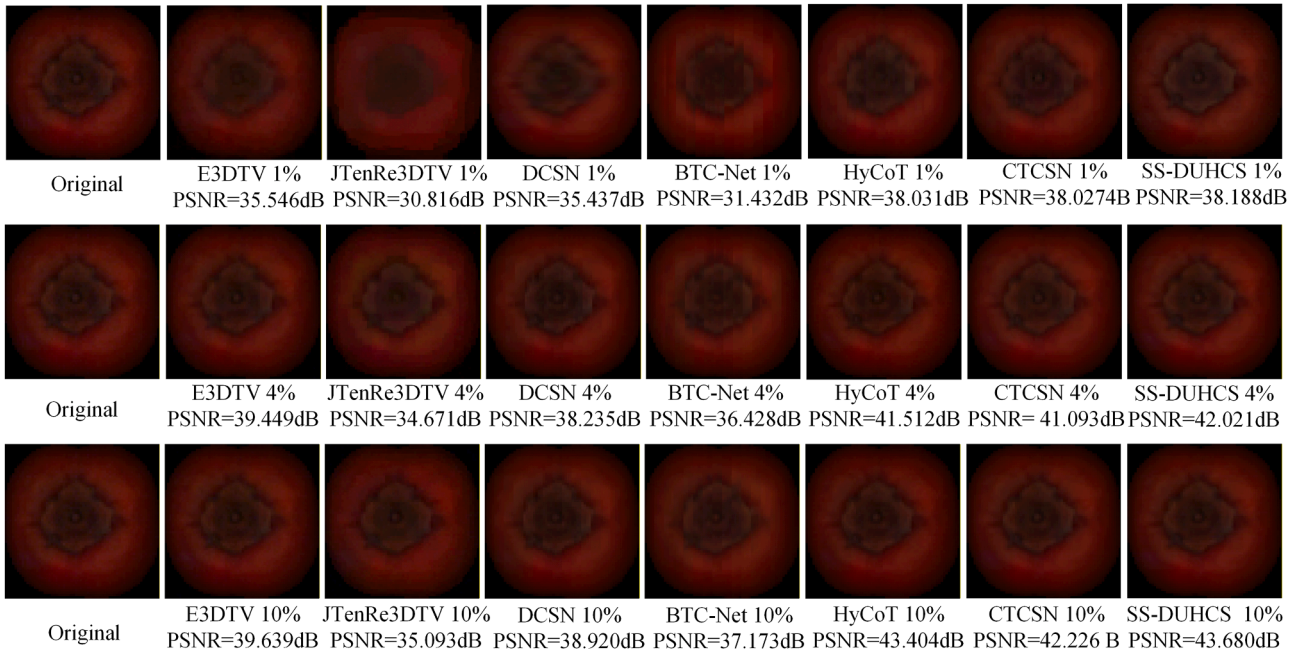
CoT, strip-based patches for DCSN and CTCN, and block-wise spatial-spectral slices for SS-DUHCS). Under this definition, SS-DUHCS requires 30.16 GFLOPs and 30.69K parameters per cube. DCSN and BTC-Net consume 13.14–46.53 GFLOPs and 19.52–54.84 GFLOPs, respectively, with over 11M parameters; CTCN needs 11.57–34.44 GFLOPs with over 7.8M parameters. HyCoT allocates 30.73–30.90 GFLOPs to a heavy Transformer encoder, while its decoder is a two-layer MLP, performing compression-decompression instead of classical CS reconstruction. Notably, SS-DUHCS maintains the same FLOPs and parameter count at 1%, 4%, and 10% sampling rates, as the initialization module reconstructs the compressed vector to the original cube shape through a fixed matrix. The subsequent unfolding stages are independent of the sampling rate. In contrast, DCSN, BTC-Net, and CTCN must adjust their convolutional encoders/decoders or attention blocks when the sampling rate changes or when the input strip size varies, explaining their rapidly increasing FLOPs at higher sampling rates. Thus, at 4% and 10%, SS-DUHCS provides higher reconstruction accuracy, similar or lower FLOPs, and two to three orders of magnitude fewer parameters, highlighting its suitability for lightweight agricultural hardware.

The average time reported in Table 2 is the mean wall-clock time required to complete the compressive measurement (encoding) and reconstruction (decoding) processes for each hyperspectral cube in the test set, from sampling to full-cube reconstruction. All deep learning

Table 2

The results of different HSI CS methods on the persimmons dataset at different sampling rates.

CS ratio (%)	Methods	PSNR (dB)	SSIM	RMSE	SAM (°)	Params (K)	FLOPs (G)	Average time(s)
1	E3DTV-7[44]	19.079	0.596	0.113	7.346	–	–	24.65
	E3DTV-70[44]	35.694	0.920	0.016	5.096	–	–	240.34
	JTenRe3DTV-7[43]	26.139	0.816	0.049	5.793	–	–	17.89
	JTenRe3DTV-70[43]	30.335	0.815	0.030	6.246	–	–	177.05
	DCSN [17]	34.935	0.933	0.018	4.430	11983.62	13.14	1.50
	BTC-Net[45]	33.676	0.886	0.021	5.636	12254.24	19.52	1.59
	HyCoT [19]	37.994	0.951	0.013	4.090	392.65	30.73	1.23
	CTCSN [20]	38.664	0.950	0.012	3.967	7863.11	11.57	1.54
	SS-DUHCS(Ours)	39.411	0.953	0.012	3.944	30.69	30.16	1.67
4	E3DTV-7[44]	24.382	0.761	0.061	6.422	–	–	26.65
	E3DTV-70[44]	39.247	0.956	0.011	3.928	–	–	270.47
	JTenRe3DTV-7[43]	26.368	0.811	0.049	6.437	–	–	24.62
	JTenRe3DTV-70[43]	34.916	0.907	0.018	5.148	–	–	190.36
	DCSN [17]	38.366	0.951	0.012	4.211	11953.60	46.34	1.51
	BTC-Net[45]	37.143	0.931	0.014	4.387	12254.24	54.72	1.68
	HyCoT [19]	41.296	0.963	0.009	3.163	404.94	30.78	1.32
	CTCSN [20]	41.167	0.964	0.009	3.080	7906.79	34.24	1.63
	SS-DUHCS(Ours)	42.169	0.978	0.008	2.865	30.69	30.16	1.85
10	E3DTV-7[44]	30.240	0.870	0.031	5.667	–	–	27.20
	E3DTV-70[44]	39.689	0.960	0.010	3.676	–	–	255.80
	JTenRe3DTV-7[43]	32.749	0.899	0.023	5.299	–	–	25.43
	JTenRe3DTV-70 [43]	35.978	0.926	0.016	5.176	–	–	237.21
	DCSN [17]	39.463	0.955	0.011	3.679	12004.54	46.53	1.68
	BTC-Net[45]	37.920	0.938	0.013	3.998	12284.77	54.84	1.76
	HyCoT [19]	42.968	0.978	0.007	2.583	433.63	30.90	1.41
	CTCSN [20]	42.092	0.970	0.008	2.691	7958.68	34.44	1.72
	SS-DUHCS(Ours)	43.649	0.980	0.007	2.454	30.69	30.16	1.94

**Fig. 5.** Comparison of RGB-visualized images for persimmon dataset using different methods at different sampling rates (1%, 4%, 10%).

methods-including DCSN, BTC-Net, HyCoT, CTCSN, and SS-DUHCS-finish within approximately 1–2 s per cube, achieving significant speedups compared to iterative methods (JTenRe3DTV and E3DTV). Although HyCoT has the shortest average time among the deep models, this is due to its heavy computation in a large Transformer-based encoder on the compression side, while the decoder is a lightweight two-layer MLP. The runtime of SS-DUHCS (1.67 s at 1% sampling) is slightly slower than DCSN (1.50 s) and comparable to BTC-Net and CTCSN, mainly due to implementation-level factors rather than algorithmic complexity. SS-DUHCS employs a block-wise processing strategy, unfolding each hyperspectral datacube row-wise into spatial-spectral slices, then

partitioning into small spatial-spectral blocks that are reconstructed on the GPU. The transfer of data back to the CPU and subsequent stitching into the full datacube introduces additional overhead, increasing wall-clock time. In contrast, models like DCSN and CTCSN process each cube monolithically on the GPU, avoiding this overhead. For HyCoT, the compression-decompression design places the computational burden on the encoder, which is problematic for agricultural push-broom HSI, where the encoder typically runs on resource-limited embedded or edge hardware. Allocating a large Transformer encoder at the acquisition side is less practical than using a lightweight measurement module, as in SS-DUHCS. The additional overhead in SS-DUHCS is primarily

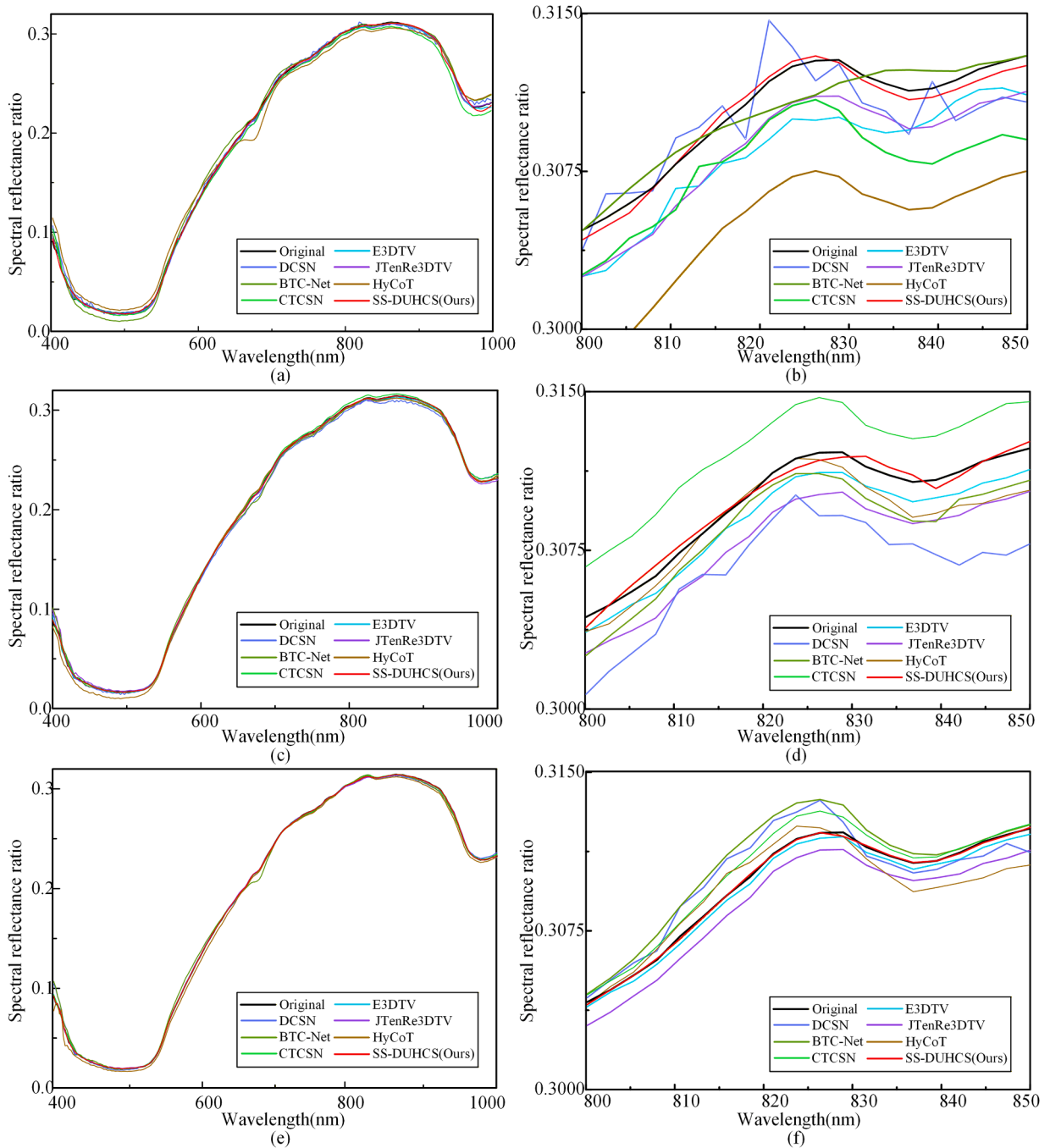


Fig. 6. Comparative analysis of average spectra and zoomed-in views for HSI cube ROI region reconstructed under varying sampling rates (1%, 4%, 10%) using different methods. (a) Average spectral curve at 1% sampling rate. (b) Local enlarged view of the average spectral curve at 1% sampling rate. (c) Average spectral curve at 4% sampling rate. (d) Local enlarged view of the average spectral curve at 4% sampling rate. (e) Average spectral curve at 10% sampling rate. (f) Local enlarged view of the average spectral curve at 10% sampling rate.

an engineering issue: migrating stitching operations to the GPU, reducing CPU-GPU data transfers, and optimizing parallel I/O can further reduce runtime while maintaining advantages in FLOPs, parameter count, and hardware compatibility at the encoder. Moreover, unlike end-to-end CNN or CNN-Transformer CS networks that require adjustments in architecture and channel configurations when input spatial size or sampling rate changes, SS-DUHCS accommodates hyperspectral cubes of varying shapes through its row-wise slicing and block-wise measurement scheme, allowing different sampling rates by simply altering the measurement matrix $\mathbf{A}$ without modifying the reconstruction network. This structural decoupling between the measurement operator and re-

construction backbone, combined with the matrix-based measurement implementation, enhances the practicality and ease of deployment of SS-DUHCS in agricultural push-broom HSI systems.

From a qualitative perspective, both spatial and spectral reconstruction quality are evaluated. For the spatial domain, RGB visualization assesses the reconstructed images. A representative hyperspectral cube from the persimmons dataset is selected, using the 114th, 57th, and 15th bands as R, G, and B, respectively. The original data and hyperspectral cubes reconstructed by different methods are visualized in RGB, as shown in Fig. 5. At a 1% sampling rate, the SS-DUHCS model produces clearer fruit contours and surface details than other methods, which

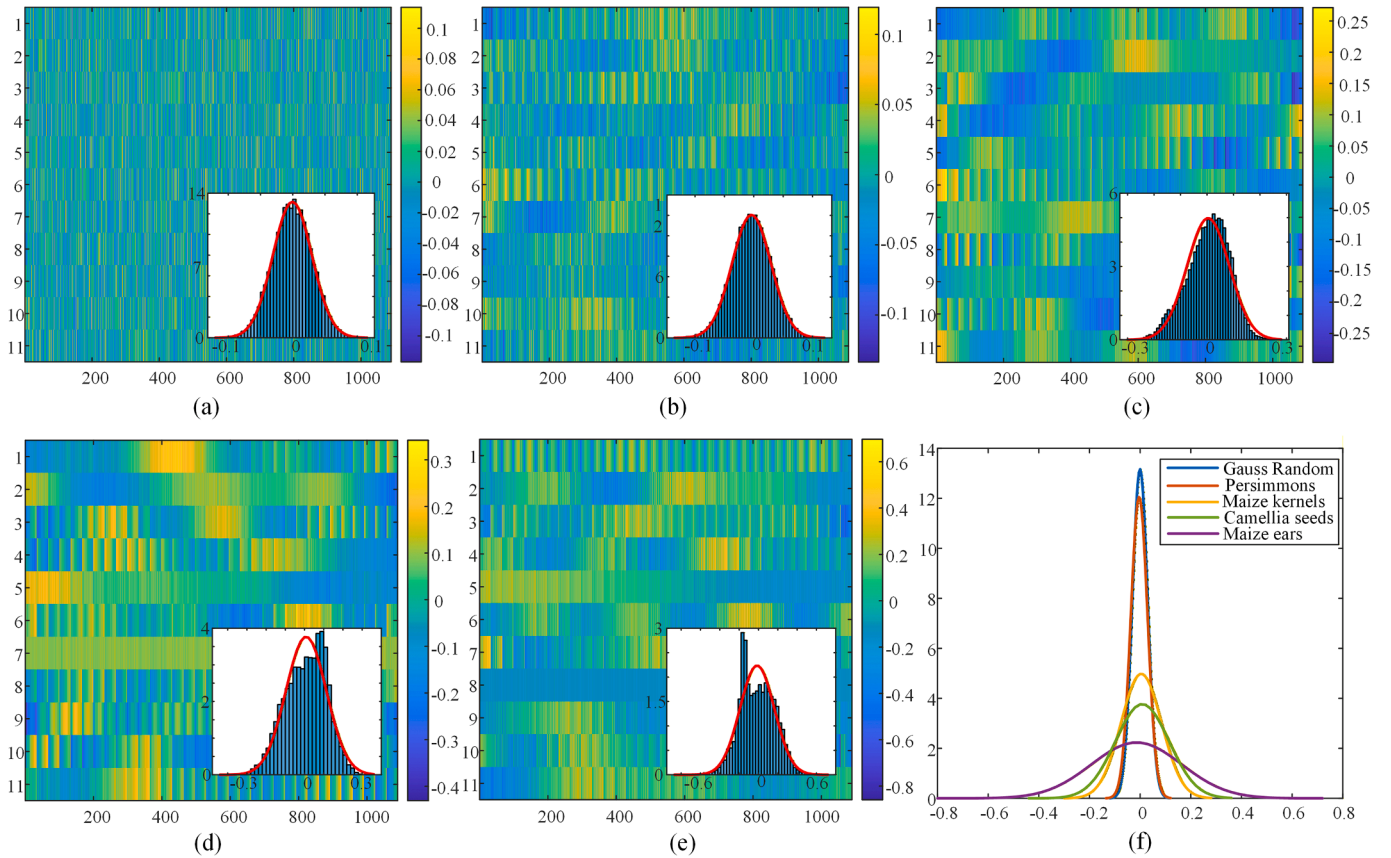


Fig. 7. Visualization of measurement matrices and corresponding distribution histograms and curves.

(a) Visualization of Gaussian random matrix with its distribution histogram. (b) Visualization and histogram of measurement matrix trained on persimmons dataset. (c) Visualization and histogram of measurement matrix trained on maize kernels dataset. (d) Visualization and histogram of measurement matrix trained on camellia oleifera seeds dataset. (e) Visualization and histogram of measurement matrix trained on maize ears dataset. (f) Comparative distribution curves of measurement matrices trained on different datasets.

exhibit blurring and checkerboard artifacts. At sampling rates of 4% and 10%, RGB images reconstructed by SS-DUHCS are nearly indistinguishable from the original image.

For agricultural product HSI, spectral information that maps to the physicochemical properties of the object holds greater application value than purely spatial information. We present the average spectral curves of a region of interest (ROI) in the persimmons dataset before and after reconstruction with different methods, as shown in Fig. 6. As the sampling rate increases, the error between the reconstructed spectral curves and the original spectrum decreases for all methods. However, at all sampling rates, the spectral curves reconstructed by SS-DUHCS consistently align most closely with the original. At a 1% sampling rate, the reconstructed spectra from the two deep learning-based methods, DCSN and BTC-Net, exhibit significant fluctuations, while the HyCoT, CTCN, JTenRe3DTV, and E3DTV methods yield smoother but still biased waveforms. At a 10% sampling rate, the curve reconstructed by the proposed method closely matches the original spectral curve, while the other methods show noticeable deviations.

3.3.2. Ablation study

To verify the contribution of each component in the proposed SS-DUHCS, ablation studies were conducted on the persimmons dataset at a 1% sampling rate. Detailed configurations and corresponding quantitative results are provided in Table 3. Baseline-I serves as the reference model, utilizing a fixed Gaussian random measurement matrix and a standard convolution-based denoiser (without trainable A and SCFRM). Baseline-II replaces standard convolutions with the proposed SCFRM

while retaining the fixed matrix (with SCFRM, without trainable A). As shown in Table 3, this configuration reduces parameters and FLOPs from 133.73K and 129.83 GFLOPs to 30.69K and 30.16 GFLOPs, respectively, achieving reductions of approximately 77% in both model size and computational cost, while also gaining +0.403 dB in PSNR and improving SSIM, RMSE, and SAM over Baseline-I. This suggests that SCFRM is not merely an attention module, but a structurally efficient refinement block that enhances feature representation and reduces the complexity of the denoiser. Baseline-III incorporates the gradient-optimized trainable measurement matrix but reverts to the standard convolutional denoiser (with trainable A, without SCFRM). This leads to a substantial improvement of +5.195 dB in PSNR and +0.083 in SSIM compared to Baseline-I, confirming that the data-driven measurement strategy effectively captures the specific physical characteristics of agricultural HSI to enhance reconstruction fidelity. Finally, SS-DUHCS integrates both the

Table 3

Results of the ablation studies on the persimmons dataset at a 1% sampling rate.

	Baseline-I	Baseline-II	Baseline-III	SS-DUHCS
SCFRM		✓		✓
Trainable A			✓	✓
PSNR (dB)↑	32.651	33.054	37.846	39.411
SSIM↑	0.862	0.865	0.945	0.953
RMSE↓	0.026	0.025	0.013	0.012
SAM (°)↓	6.571	6.365	4.371	3.944
Params (K)	133.73	30.69	133.73	30.69
FLOPs (G)	129.83	30.16	129.83	30.16

Table 4

Cross-dataset generalization performance of models trained with 1% sampling rate across agricultural products HSI datasets.

Training Set	Measure	Testing Set				Averaged
		Persimmons	Maize kernels	Camellia seeds	Maize ears	
Persimmons	PSNR↑	39.411	41.554	35.582	39.217	38.941
	SSIM↑	0.953	0.952	0.899	0.945	0.937
	RMSE↓	0.012	0.008	0.017	0.011	0.012
	SAM↓	3.944	1.682	3.382	2.023	2.758
Maize kernels	PSNR↑	36.822	42.627	35.271	38.836	38.389
	SSIM↑	0.935	0.964	0.903	0.947	0.937
	RMSE↓	0.015	0.007	0.018	0.011	0.013
	SAM↓	4.856	1.538	3.456	2.107	2.989
Camellia seeds	PSNR↑	35.316	42.275	36.600	37.465	37.914
	SSIM↑	0.914	0.961	0.909	0.923	0.927
	RMSE↓	0.018	0.008	0.015	0.013	0.014
	SAM↓	5.398	1.590	3.081	2.189	3.064
Maize ears	PSNR↑	36.277	42.17	36.21	40.934	38.898
	SSIM↑	0.927	0.961	0.906	0.956	0.937
	RMSE↓	0.016	0.008	0.016	0.009	0.012
	SAM↓	4.888	1.612	3.213	1.806	2.880
Averaged	PSNR↑	36.956	42.157	35.916	39.113	38.535
	SSIM↑	0.932	0.96	0.904	0.943	0.935
	RMSE↓	0.015	0.008	0.016	0.011	0.013
	SAM↓	4.772	1.605	3.283	2.031	2.923
E3DTV	PSNR↑	35.694	41.594	34.969	37.049	37.326
	SSIM↑	0.920	0.944	0.875	0.909	0.912
	RMSE↓	0.016	0.008	0.018	0.014	0.014
	SAM↓	5.096	1.711	3.615	3.211	3.408
JTenRe3DTV	PSNR↑	30.335	39.106	32.814	32.28	33.634
	SSIM↑	0.815	0.936	0.855	0.864	0.867
	RMSE↓	0.030	0.011	0.023	0.024	0.022
	SAM↓	6.246	2.016	4.547	4.402	4.303

trainable A and SCFRM, achieving the best overall performance with relative improvements of 20.7% in PSNR, 10.5% in SSIM, a 53.8% reduction in RMSE, and a 40.0% reduction in SAM compared with Baseline-I, while maintaining the same lightweight parameter count (30.69K) and computational complexity (30.16 GFLOPs) as Baseline-II. These results demonstrate that synergistic integration of spatial-spectral deep unfolding and adaptive compressive sensing is essential for achieving high-fidelity, lightweight agricultural HSI reconstruction.

3.3.3. Cross-dataset evaluation

Given the substantial feature disparities often observed among HSI datasets of agricultural products, this study evaluates the applicability and generalization of the proposed model across diverse datasets. The model is trained separately on the persimmons, maize ears, camellia oleifera seeds, and maize kernels datasets, with each serving as a separate test set. Experiments are conducted at a consistent 1% sampling rate, and reconstruction results are reported in Table 4. Performance is assessed using four key metrics: PSNR, SSIM, RMSE, and SAM. Reconstruction accuracy significantly improves when the training and test datasets originate from the same source. The average reconstructed PSNR exceeds 35 dB across all datasets, demonstrating the model's ability to accurately reconstruct the original data and affirming its robust generalization capabilities. The maize kernels dataset yields the highest reconstruction quality, with average PSNR values surpassing 41.5 dB and SAM values below 1.7° when tested with models trained on different datasets, reflecting its stronger signal correlation and sparsity, which facilitate reconstruction. In contrast, the camellia oleifera seeds dataset exhibits the poorest reconstruction performance, with average PSNR values below 37 dB across all trained models. However, performance variation across different training datasets is relatively small, indicating that this dataset contains more complex spectral and structural char-

acteristics. The persimmons dataset, acquired using a different hyperspectral sensor, experiences significant performance degradation when tested with models trained on the other datasets. To objectively quantify the model's reconstruction performance across datasets, average performance metrics are computed for each test set. For comparison, the E3DTV and JTenRe3DTV methods are evaluated on the same datasets; as iterative methods, they require no training and are tested directly. The deep learning methods DCSN, BTC-Net, CTCN, and HyCoT involve dataset-specific training and are not included in this cross-dataset setting. Across all four datasets, the average reconstruction performance of the proposed model, when trained on different datasets, consistently surpasses that of the two methods.

This model utilizes gradient optimization to refine the measurement matrix during training, replacing the traditional fixed matrix typically used in denoising models. The trained matrix effectively learns measurement-specific degradation patterns across diverse datasets, leading to improved reconstruction performance at the same sampling rate. Fig. 7 visualizes the measurement matrices and presents their distribution histograms and fitted curves. Compared to the initial Gaussian random matrix, the measurement matrices trained on different datasets exhibit smoother spatial distributions and greater statistical variance. This indicates that training the measurement matrix enables the adaptive learning of dataset-specific features, facilitating more effective data representation and capturing degradation patterns that extend beyond those modeled by a Gaussian distribution. The measurement matrix trained on the persimmons dataset exhibits notable structural differences from those trained on the other three datasets, primarily due to variations in the HSI sensors used for data acquisition. Additionally, since the real-world size represented by each pixel is larger in the latter datasets than in the persimmons dataset, their trained measurement matrices demonstrate smoother spatial distributions. Among the datasets acquired using the same HSI sensor, the

sequential increase in spatial object size across maize kernels, camellia oleifera seeds, and maize ears correlates with progressively greater dispersion, reflected in the increasing variance of the corresponding trained matrices.

4. Conclusion

To the best of our knowledge, the proposed SS-DUHCS model specifically tailored for agricultural applications represents the first attempt to integrate deep learning into agricultural products HSI compressed sensing. The proposed model balances compressive-reconstruction performance, speed, and interpretability, achieving rapid and high-quality reconstruction at low compressive sampling rates, thereby addressing data transmission and storage challenges in HSI-based agricultural applications.

(1) During the compressive stage, the approach leverages the spatial and spectral properties of agricultural products' HSI. It employs a block compressive strategy that unfolds the data row-wise into spatial-spectral slices, enhancing the sparse representation and establishing a solid foundation for compressive reconstruction. A gradient optimization technique incorporates the hyperspectral characteristics of various agricultural products into the measurement matrix, improving the model's adaptability and generalization. This matrix-based measurement design enables SS-DUHCS to accommodate hyperspectral cubes of varying spatial sizes without altering the reconstruction network. Different sampling rates can be achieved by modifying only the measurement matrix, facilitating hardware implementation on push-broom line-scan systems.

(2) In the reconstruction stage, a DUN method achieves rapid reconstruction while maintaining physical interpretability. The SCFRM further enhances reconstruction quality by extracting salient features, eliminating redundancy, and reducing model parameters and computational complexity.

(3) Experimental validation used four datasets, both open-source and self-constructed. Results were analyzed qualitatively and quantitatively. The proposed method consistently outperformed existing HSI CS techniques at compressive sampling rates of 1%, 4%, and 10%.

This work focuses on the HSI of agricultural products, utilizing the SS-DUHCS framework, which does not depend on agriculture-specific priors hard-coded into the model architecture. It learns task-relevant spatial-spectral representations and an adaptive measurement matrix from data in a model-inspired manner. Therefore, the same measurement-reconstruction pipeline can be applied to other HSI domains, such as remote sensing and biomedical HSI, by retraining the measurement matrix and the unfolding-based reconstruction network on domain-specific datasets. Thus, SS-DUHCS provides a general framework for CS and reconstruction in push-broom and line-scan HSI systems. Our experiments on four distinct agricultural datasets serve as a representative application case. Future research will investigate compressive methods with variable sampling rates to reduce data volume while enhancing reconstruction performance.

CRedit authorship contribution statement

Zhenyu Han: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Xuan Liu:** Methodology, Software, Supervision; **Guanghui Yang:** Formal analysis, Validation, Writing – original draft; **Meipan Wang:** Formal analysis, Writing – original draft; **Jian Li:** Formal analysis, Project administration, Resources, Supervision; **Yongli Zhang:** Funding acquisition, Project administration, Resources, Supervision; **Tailin Han:** Project administration, Resources, Supervision, Validation.

Data availability

Part of the data used in this study were obtained from the dataset reported in [37]. The remaining data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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